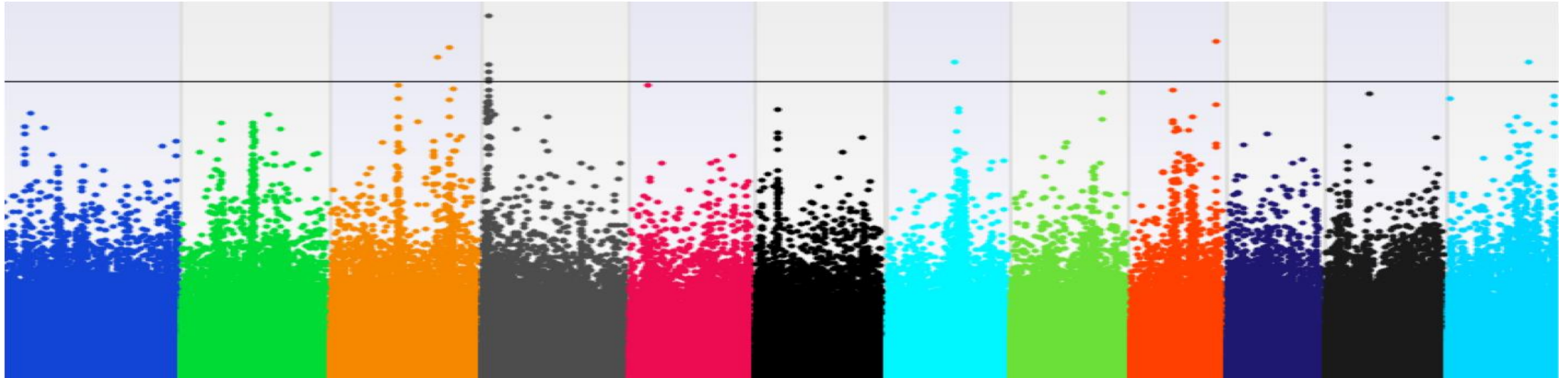


QTL/Gene Discovery using GWAS

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BIOTEC, NSTDA



Outline

- Define genetic association studies
- Linkage disequilibrium (LD)
- Overview of essential components of a GWAS analysis
- Practice on GWAS using TASSEL

Association Analysis

Searching for genotype-phenotype correlations among unrelated individuals.

Genotyping						
Individua 1	A	C	G	A	G	1.3 m
Individua 2	A	C	G	A	T	1.4 m
Individua 3	A	T	A	A	G	1.5 m
Individua 4	C	T	A	G	T	1.8 m
Individua 5	A	C	G	G	T	2.0 m
Individua 6	A	T	G	G	G	2.1 m
A/C T/C G/A A/G G/T						

Phenotyping



Association mapping

- Studying the genetics of natural variation
- Understanding the genetic architecture of traits of ecological and agricultural importance
- Identifying the genomic regions that control genetic variation
- Test association at many variants instead of some and hypothesis-free instead of hypothesis-driven.

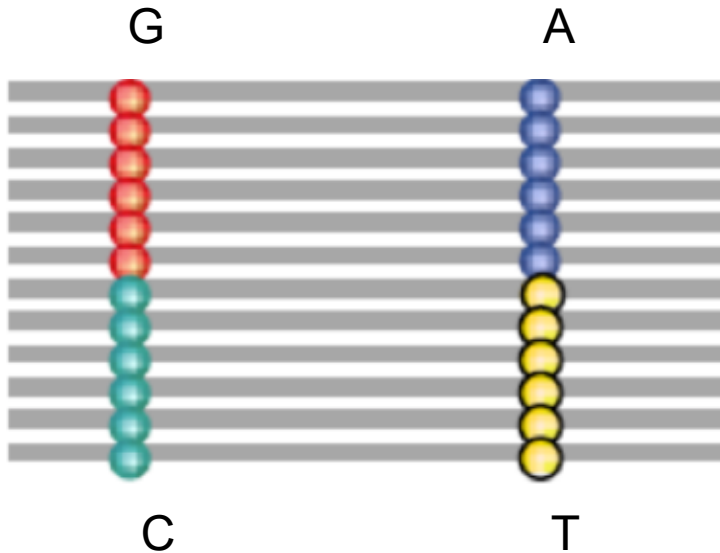
The aim of genome-wide association studies is to determine relationships between phenotypes and genotypes.

What is GWAS?

- Genome-Wide Association Study (GWAS) – An examination of many **common genetic variants** in different individuals to see if any variant is associated with a trait.
- Under the assumption that is “**common genetic variation explains common phenotypic variation**”

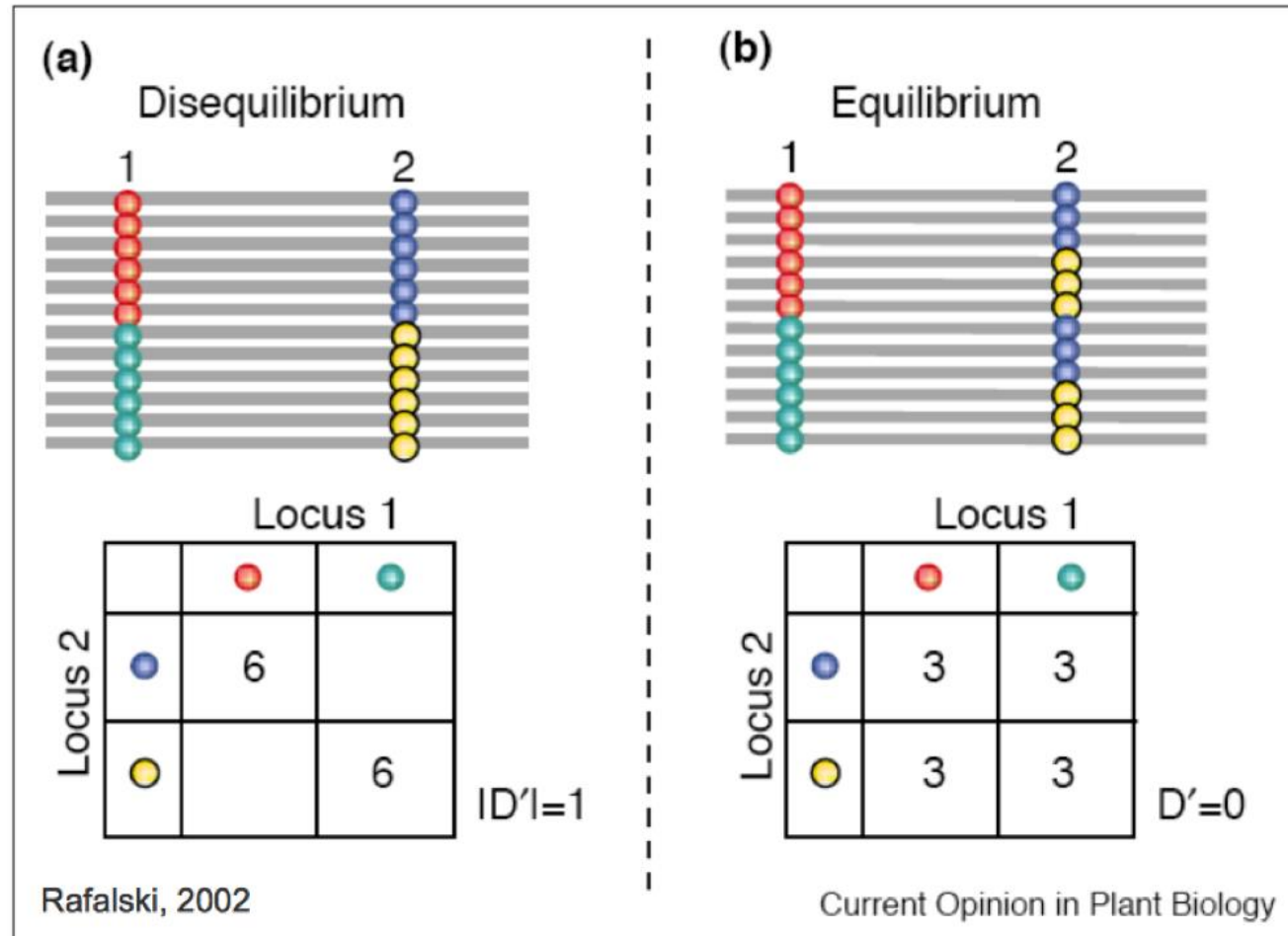
Linkage Disequilibrium (LD)

- The non-random association of alleles at linked loci
- A measure of the tendency of some alleles to be inherited together on haplotypes descended from ancestral chromosomes



- The allele G and A (C and T) are in perfect LD.
- If we genotype the first SNP, we know what the alleles are at the second SNP.

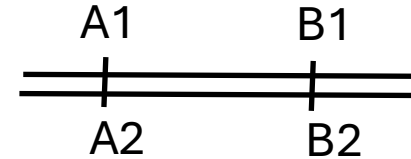
A Conceptual View of LD



r^2 Calculation

- Consider a situation where the allele frequencies are

- $f(A1) = f(A2) = f(B1) = f(B2) = 0.5$



- The haplotype frequencies are:

- $f(A1B1) = 0.1$

- $f(A1B2) = 0.4$

- $f(A2B1) = 0.4$

- $f(A2B2) = 0.1$

$$D = P_{A1B1} - P_{A1}P_{B1} = P_{A1B1}P_{A2B2} - P_{A1B2}P_{A2B1}$$

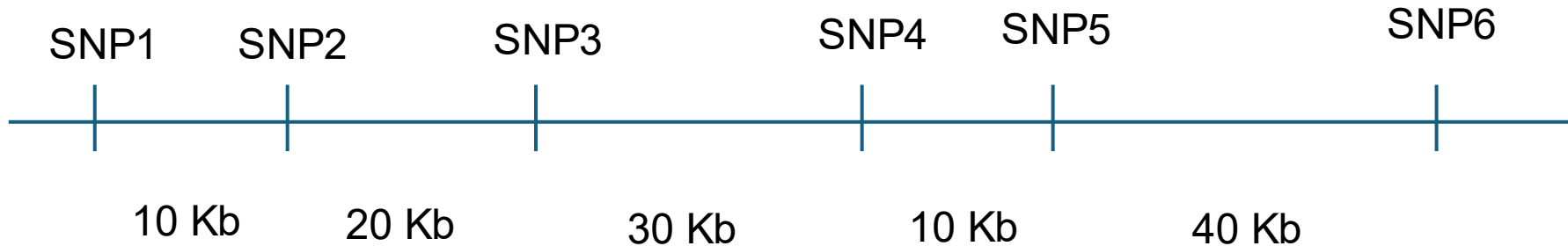
- The $D = (0.1 \times 0.1) - (0.4 \times 0.4) = -0.15$

- $D^2 = 0.0225$

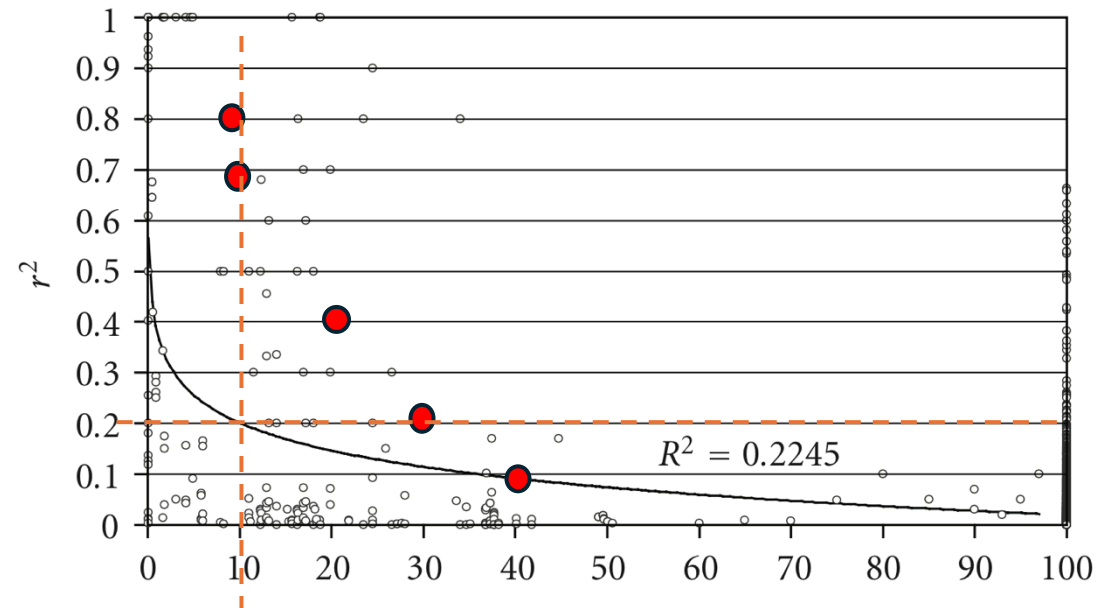
- $r^2 = 0.0225 / (0.5 \times 0.5 \times 0.5 \times 0.5) = 0.36$

$$r^2 = \frac{D^2}{f(A1) \times f(A2) \times f(B1) \times f(B2)}$$

LD Decay Estimation

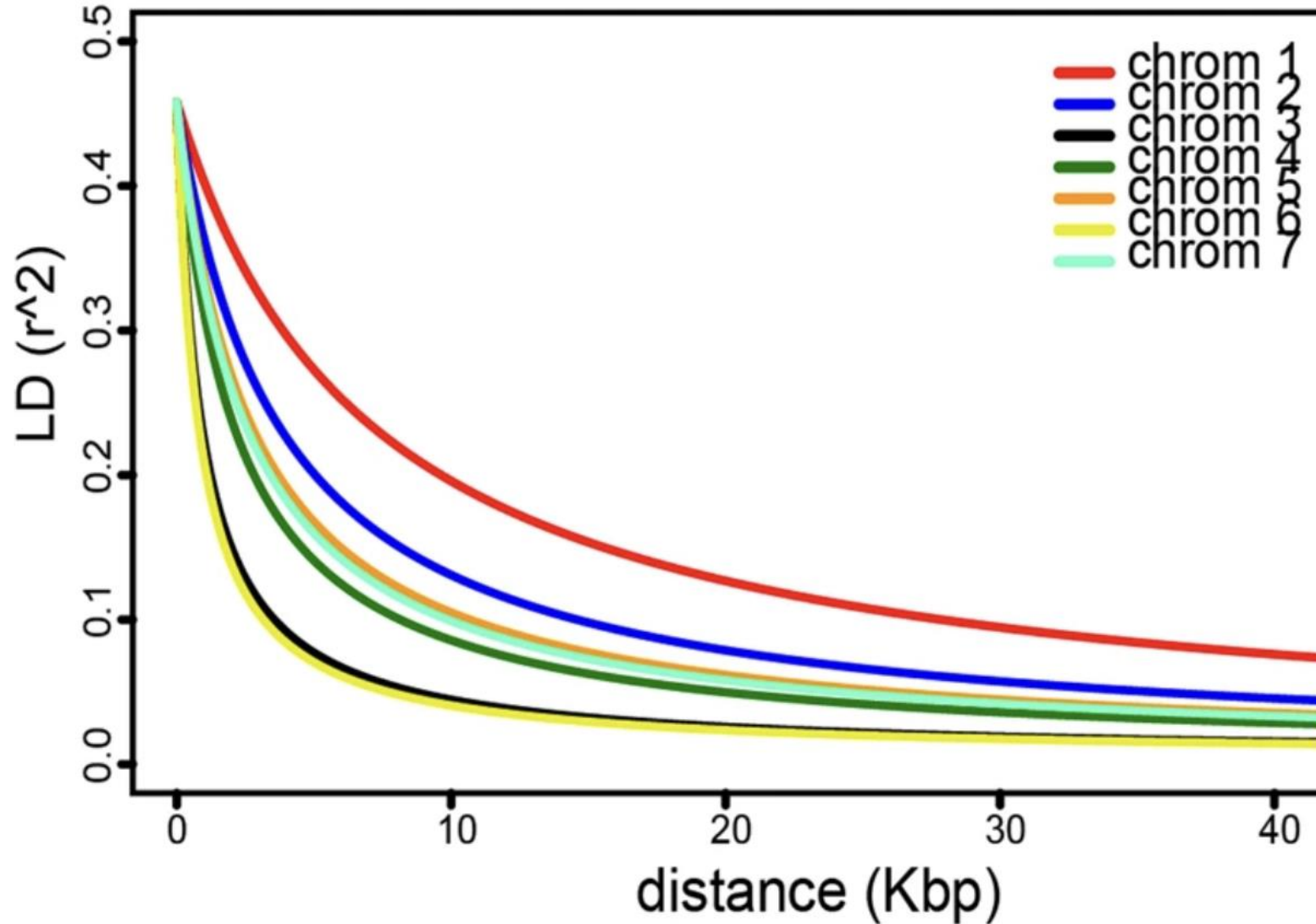


- Calculate r^2 for each pair of adjacent SNPs
 - SNP1- SNP2 (10 kb) $\rightarrow r^2 = 0.8$
 - SNP2- SNP3 (20 kb) $\rightarrow r^2 = 0.4$
 - SNP3 - SNP4 (30 kb) $\rightarrow r^2 = 0.2$
 - SNP4 - SNP5 (10 kb) $\rightarrow r^2 = 0.7$
 - SNP5 - SNP6 (40 kb) $\rightarrow r^2 = 0.1$

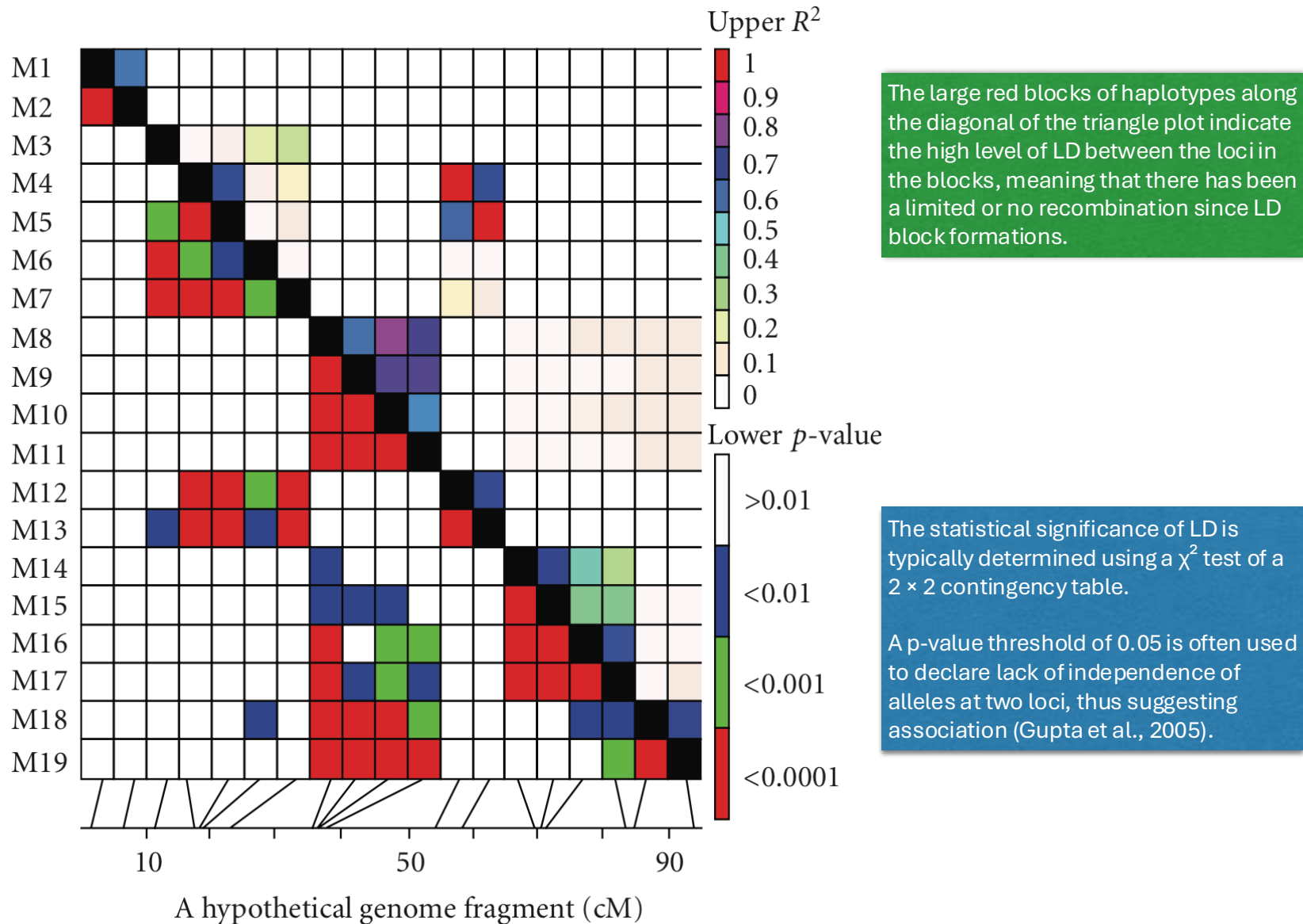


LD decay (r^2) cut-off = 0.1 or 0.2

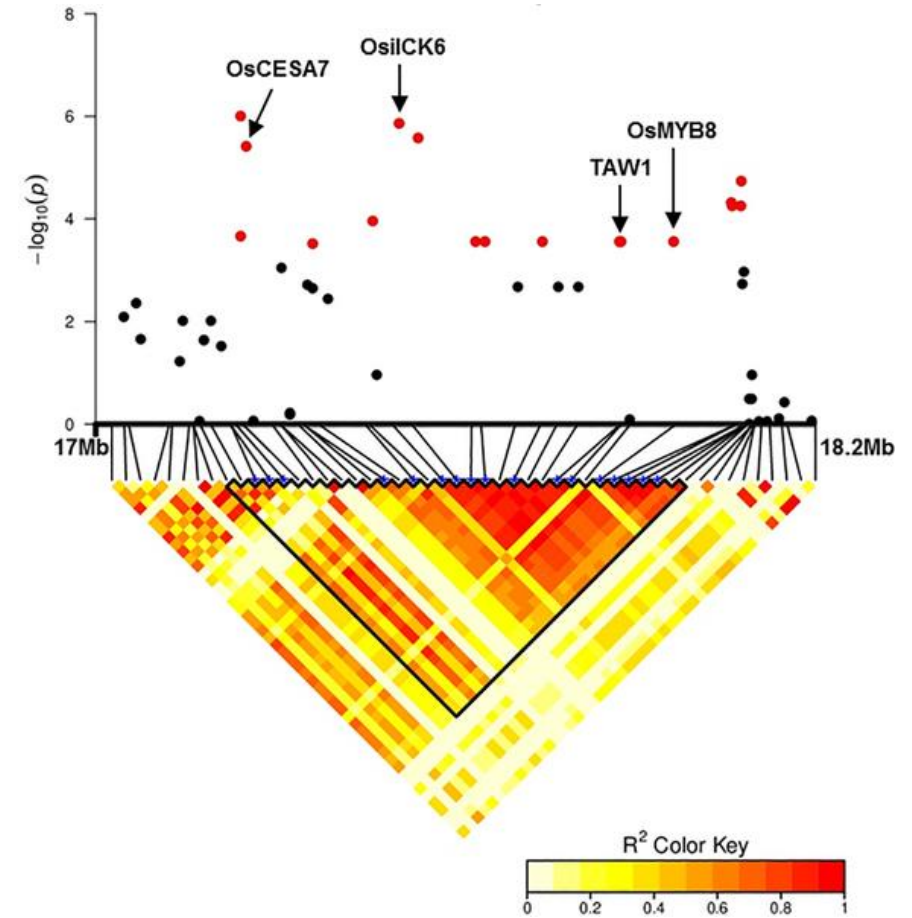
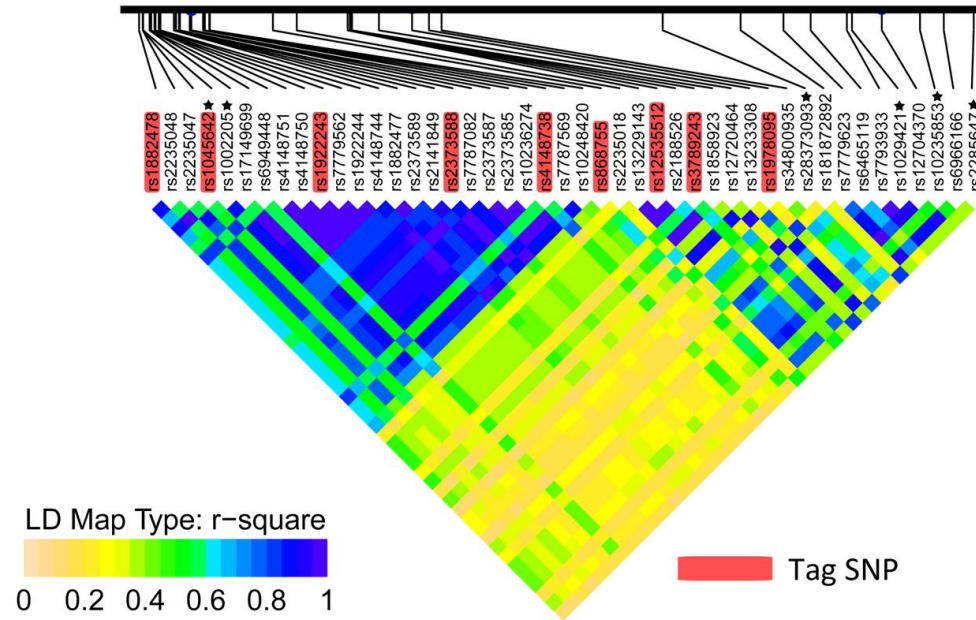
Chromosome-wise LD-decay plots



LD Matrix or LD Heat Map

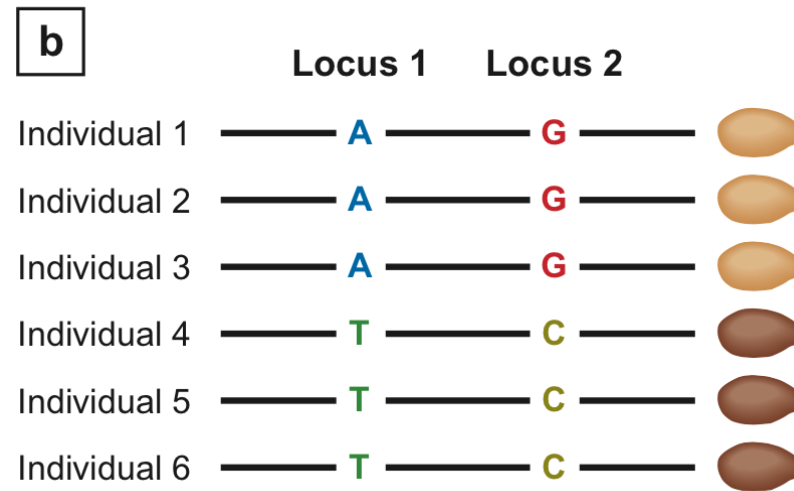


Local LD Heatmap



LD causes association that enable us to perform GWAS

- LD is non-random association of alleles between genetic loci.
- The key to association mapping is the LD between functional loci and markers that are physically linked.

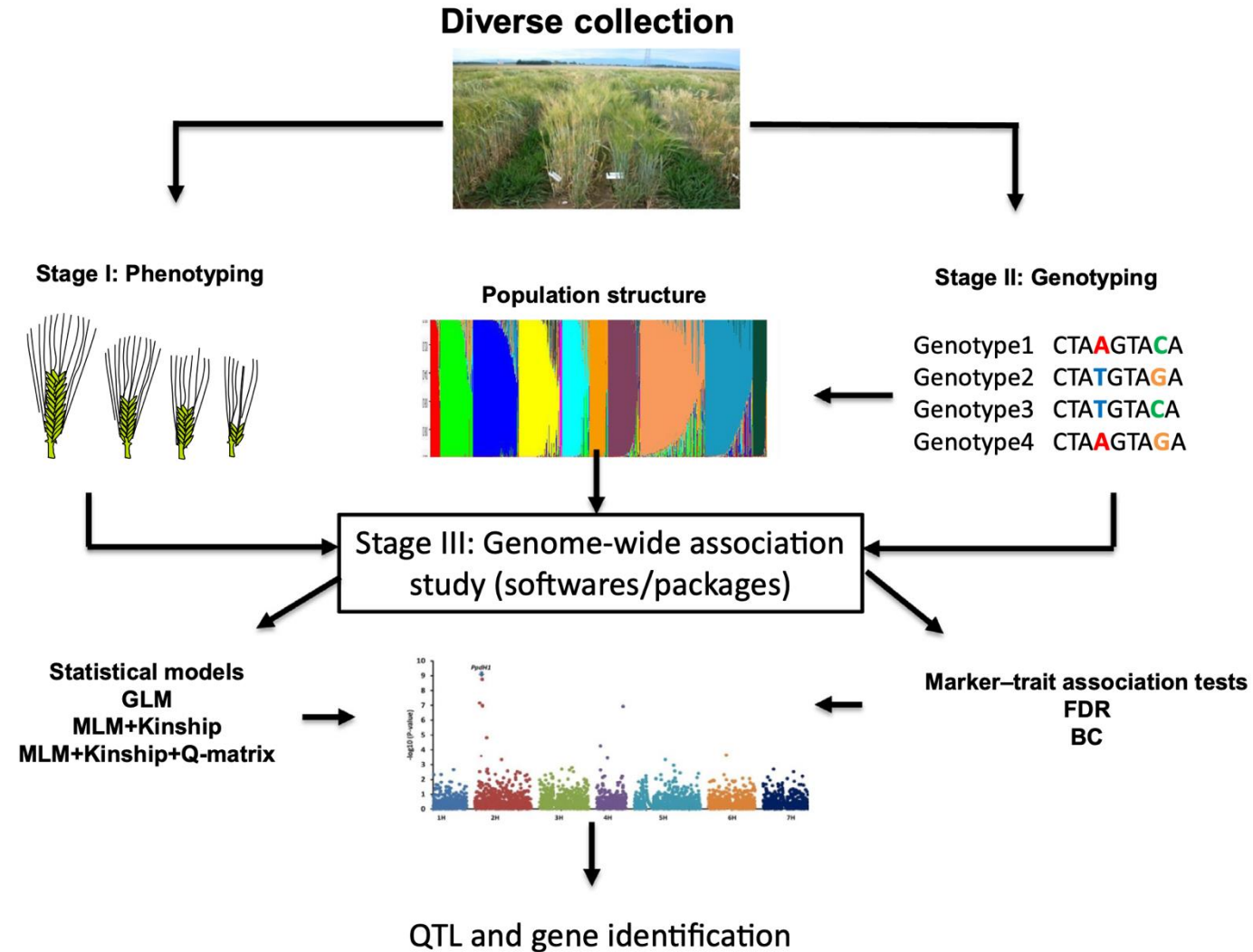


Significant association (covariance) between Locus 1 and Locus 2 with the seed color phenotype

GWAS approach

1. Collection of germplasm with wide coverage of genetic diversity
2. Phenotyping the population samples for trait of interest under different environment
3. Genotyping of mapping population with molecular markers
4. Quantification of LD for chosen population genome using molecular marker data
5. Assessment of the population structure and kinship
6. Correlation of phenotypic and genotypic data with statistical approach

The Most Important Three Stages of GWAS Experiments



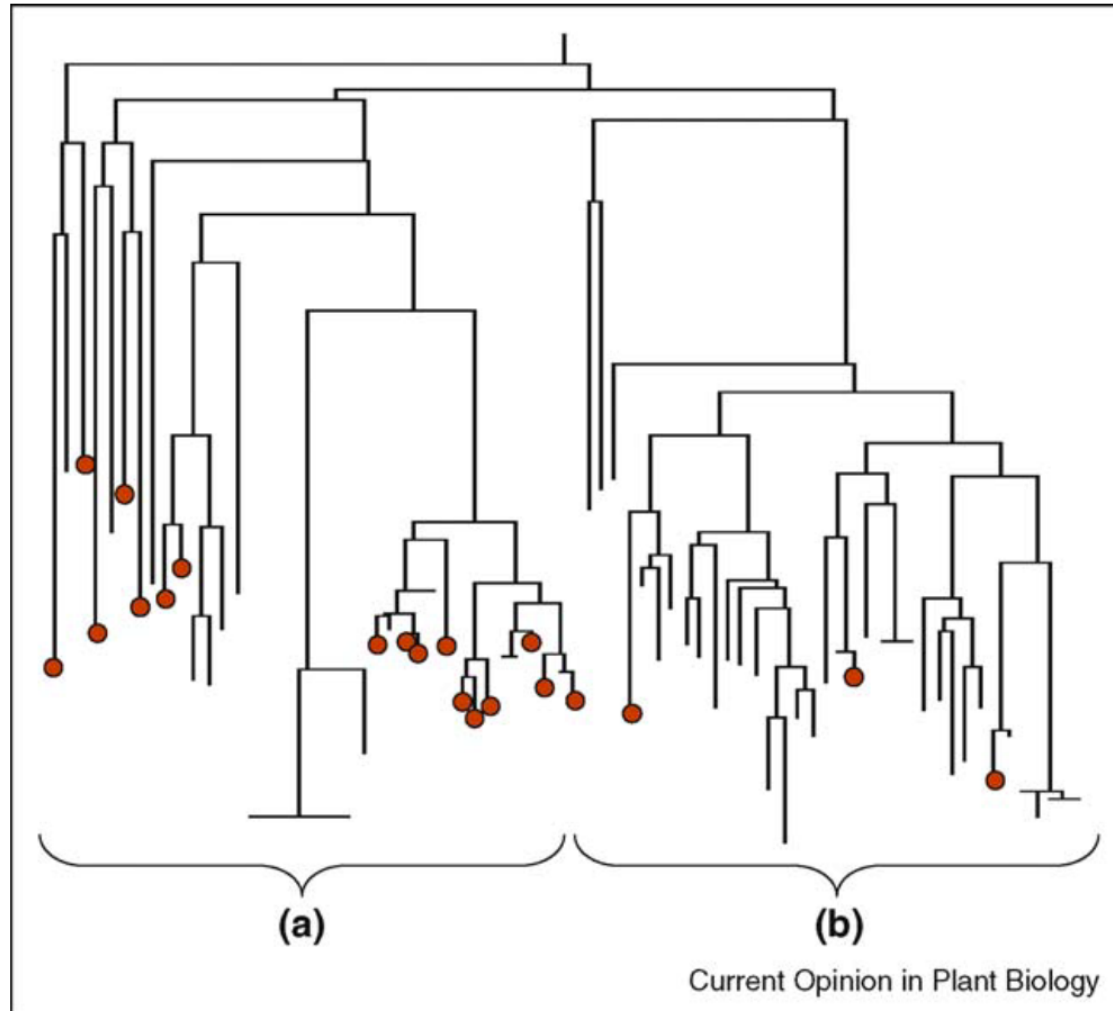
Requirements for GWAS

- Genetic diversity panel (association mapping population)
- Genotyping data
 - SNPs
- Phenotypic (trait) data

Association Mapping Populations

- (1) A large random sample from a natural population
- (2) A germplasm core collection
- (3) A collection of breeding lines including cultivars
- (4) A population derived from multiparent crosses

Confounding of Association Mapping by Population Structure



- Red dots (i.e. disease resistance) is common in subgroup A but rare in subgroup B.
- Any markers with significantly uneven allele distribution between the two subgroups will be positive in association test, irrespective of their genomic locations.

The statistical model of quantitative genetics

- The model adjust the phenotypic value to a genetic and an environmental component

Phenotypic value = Mean + Genotypic value + Environmental deviation

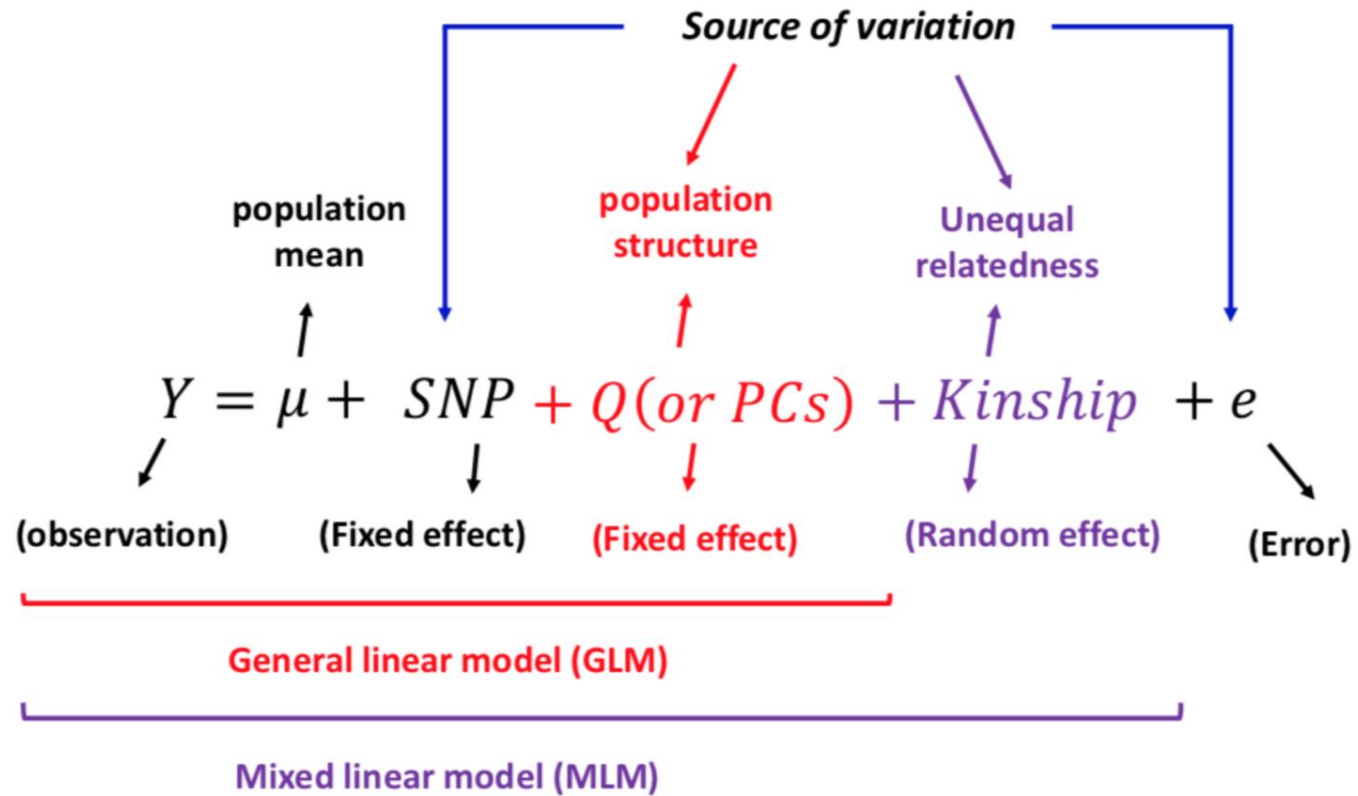
$$P = m + G + E$$

- E.g. Height

$$1.80 = 1.70 + 0.15 - 0.05$$

Phenotype = Mean + Genotypic value – Environmental deviation

GWAS models



How is marker-trait tested?

- Testing full model over reduced model to see if SNP has significant effect on trait

$$Y = \mu + \textcolor{red}{SNP} + Q + e \quad (\text{Full model})$$

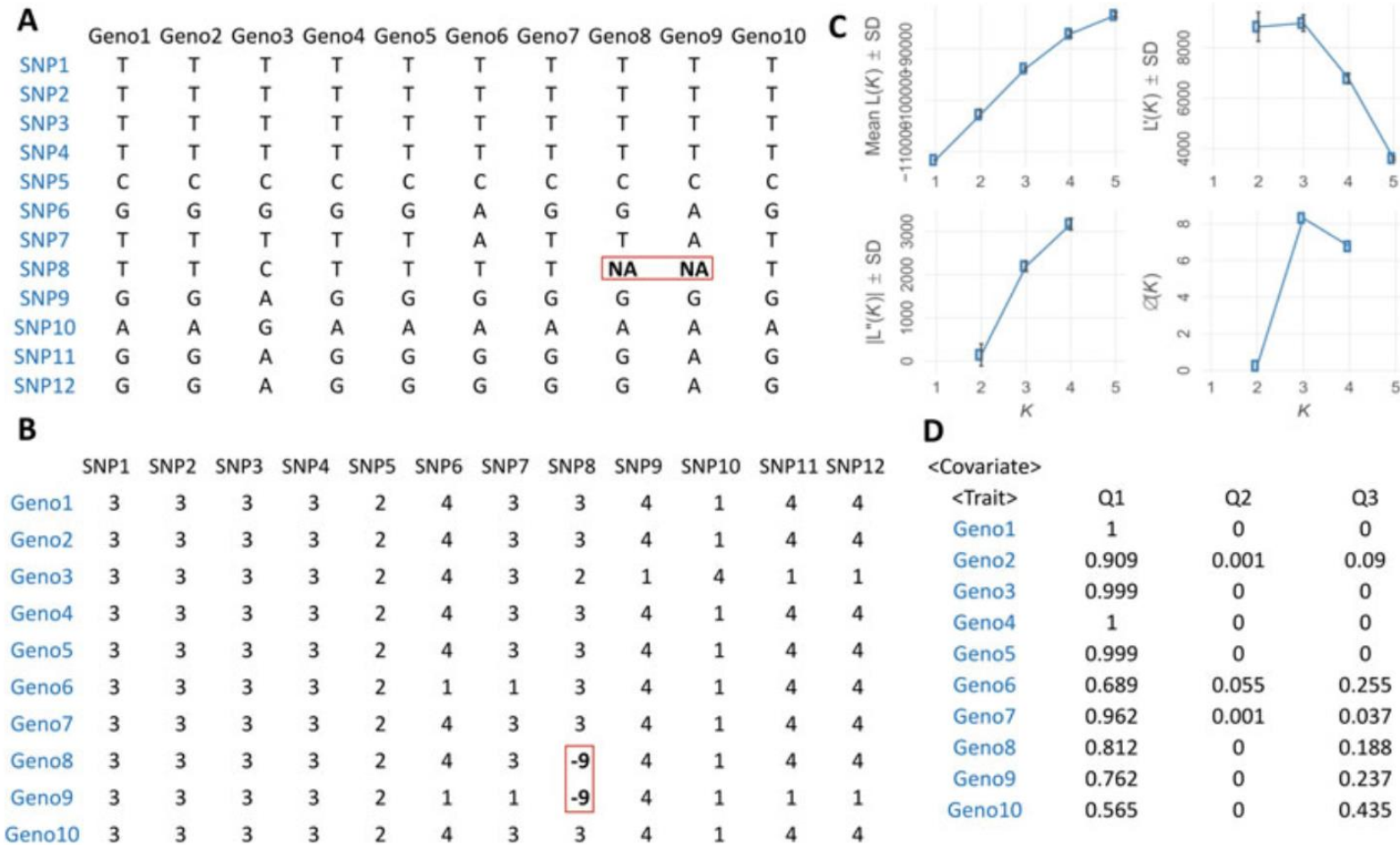
$$Y = \mu + Q + e \quad (\text{Reduced model})$$

$$\frac{\text{Full model}}{\text{Reduced model}}, \quad P \text{ value}$$

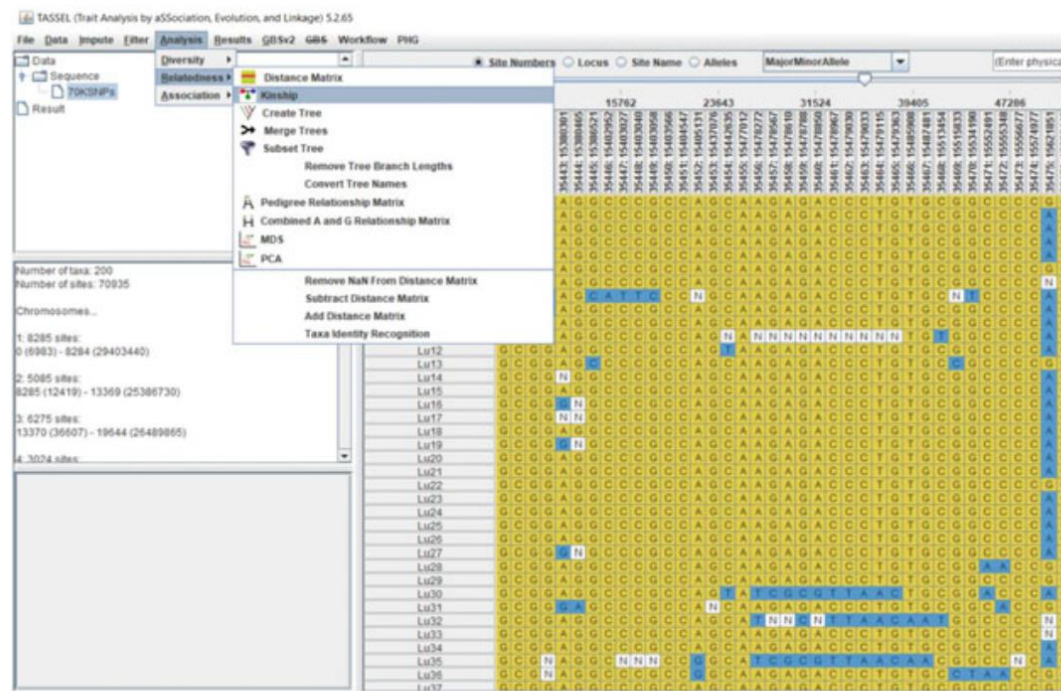
Mixed Linear Model (MLM)

- Q matrix (STRUCTURE) - fixed effect
 - can be replaced by PCA (P matrix)
- K matrix - random effect
 - estimates the identity by descent (IBD) by adjusting the probability of identity by state (IBS) between two individuals

Q-matrix



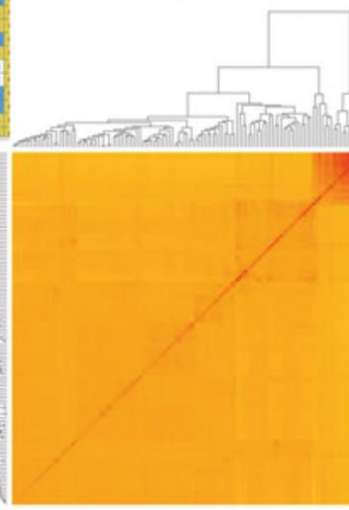
K-matrix



Kinship heatmap

Kinship matrix

	Geno1	Geno2	Geno3	Geno4	Geno5	Geno6	Geno7	Geno8	Geno9	Geno10
Geno1	1.9010	0.0235	-0.0924	0.2808	-0.0037	0.0351	-0.0045	0.0433	-0.0088	0.0332
Geno2	0.0235	1.3399	-0.1901	0.2485	0.2972	0.1102	0.1171	0.1926	0.2173	0.1563
Geno3	-0.0924	-0.1901	2.9976	-0.1699	-0.1797	-0.0997	-0.0459	-0.1560	-0.1082	0.0353
Geno4	0.2808	0.2485	-0.1699	1.1288	0.1854	0.1434	0.0837	0.2144	0.1001	0.0631
Geno5	-0.0037	0.2972	-0.1797	0.1854	0.6834	0.1341	0.0584	0.2090	0.1657	-0.0071
Geno6	0.0351	0.1102	-0.0997	0.1434	0.1341	0.8290	0.0258	0.1645	0.0123	-0.0238
Geno7	-0.0045	0.1171	-0.0459	0.0837	0.0584	0.0258	1.7781	0.0605	0.1195	0.1043
Geno8	0.0433	0.1926	-0.1560	0.2144	0.2090	0.1645	0.0605	0.7153	0.0678	0.0031
Geno9	-0.0088	0.2173	-0.1082	0.1001	0.1657	0.0123	0.1195	0.0678	2.0379	0.2765
Geno10	0.0332	0.1563	0.0353	0.0631	-0.0071	-0.0238	0.1043	0.0031	0.2765	1.3441



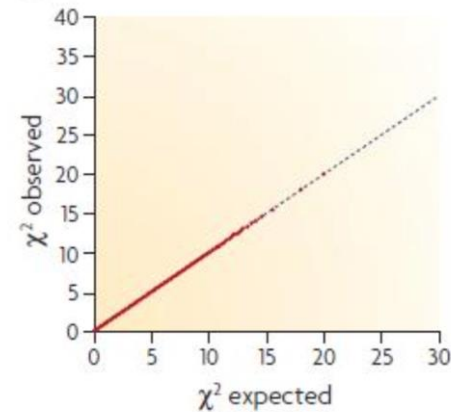
Software for Association Mapping

- TASSEL 5.0 (Trait Analysis by Association, Evaluation and Linkage) - <https://tassel.bitbucket.io>
 - GLM, MLM, compressed MLM, and P3D approaches for marker-trait association analysis
- GAPIT 3 (Genome Association and Prediction Integrated Tool) - <https://zzlab.net/GAPIT/>
 - R package
 - GLM, MLM, compressed MLM (CMLM), enriched CMLM (ECMLM), multiple locus mixed linear model (MLMM), settlement of MLM under progressively exclusive relationship (SUPER), fixed and random model circulating probability unification (FarmCPU), and Bayesian-information and linkage-disequilibrium iteratively nested keyway (BLINK)

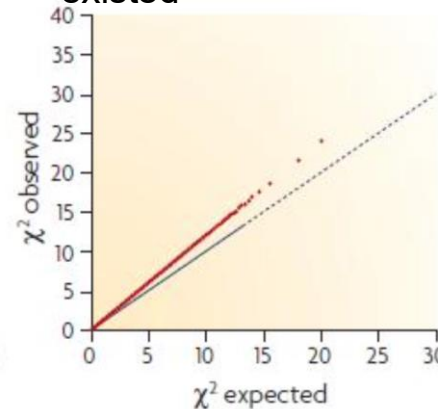
Quantile-quantile plots (Q-Q plots)

- For model checking.
- Compare the quantiles of the empirical test statistic distribution to assumed null distribution.
- Sort test statistic, then plot test statistic (Y-axis) against theoretical one (X-axis).
- Deviation of plots from the diagonal indicates inflation or deflation of test statistics.

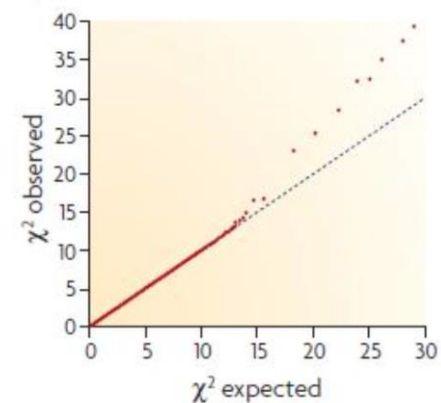
no association



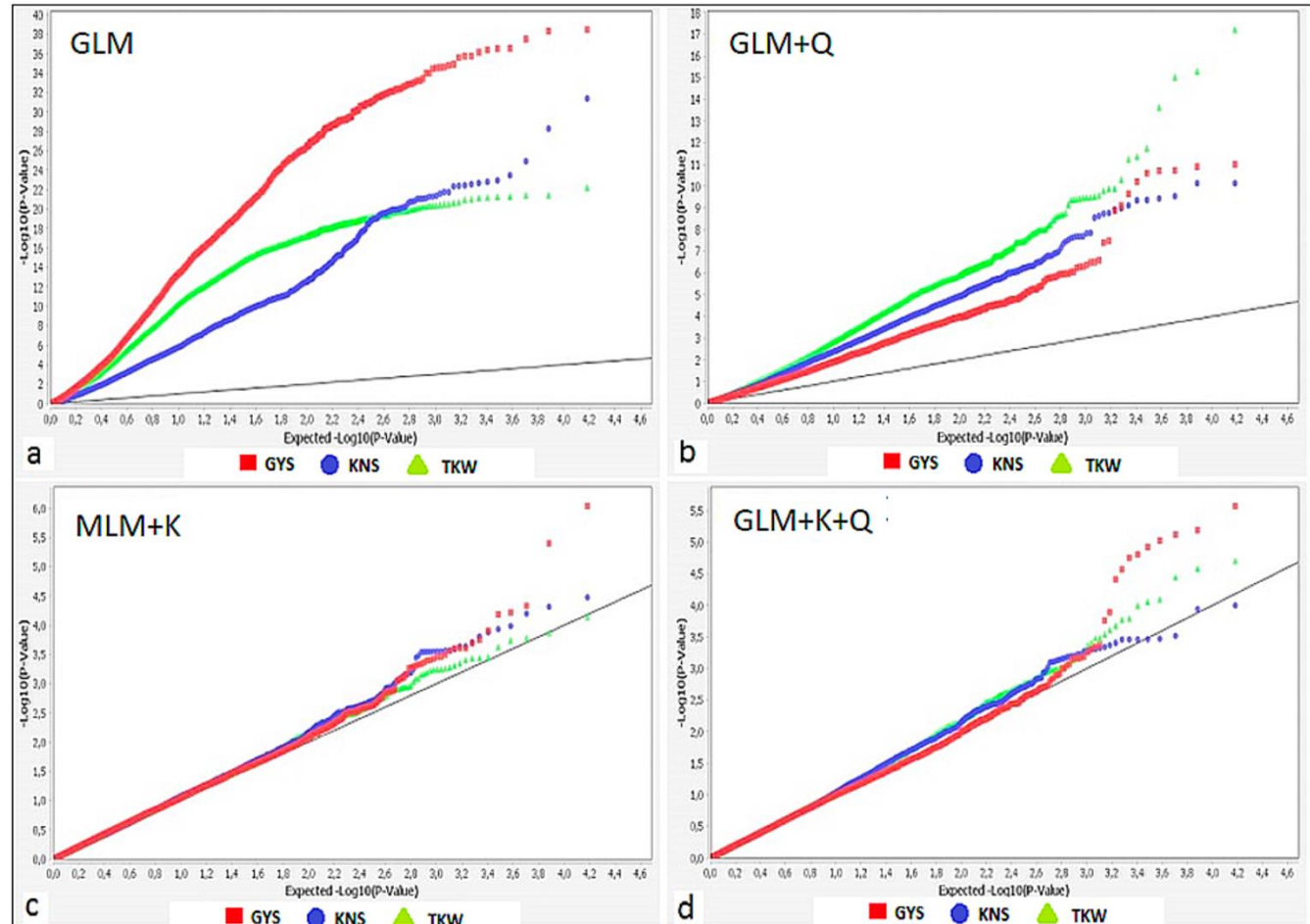
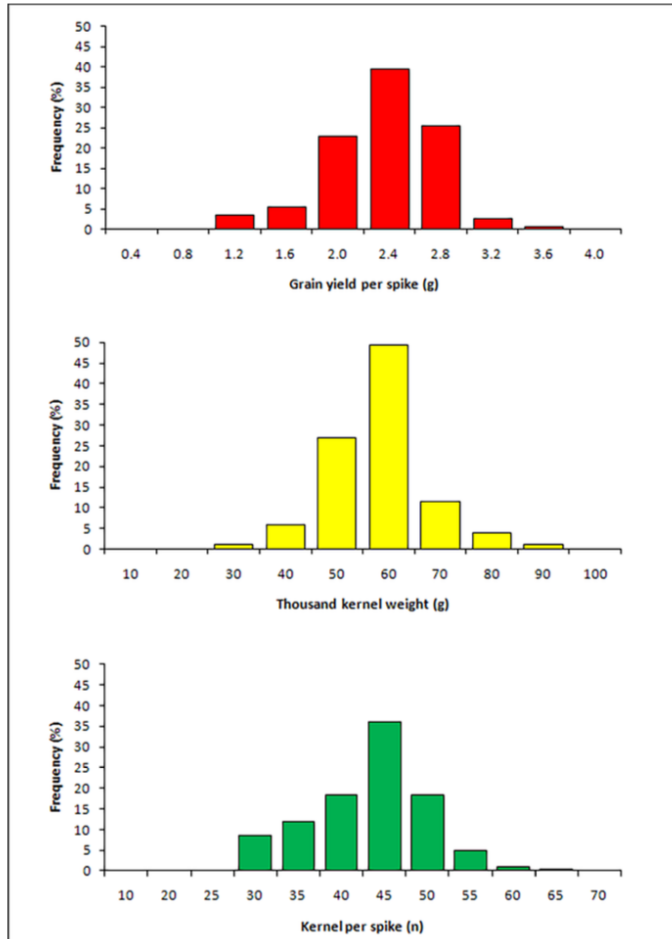
inflated,
population structure
existed



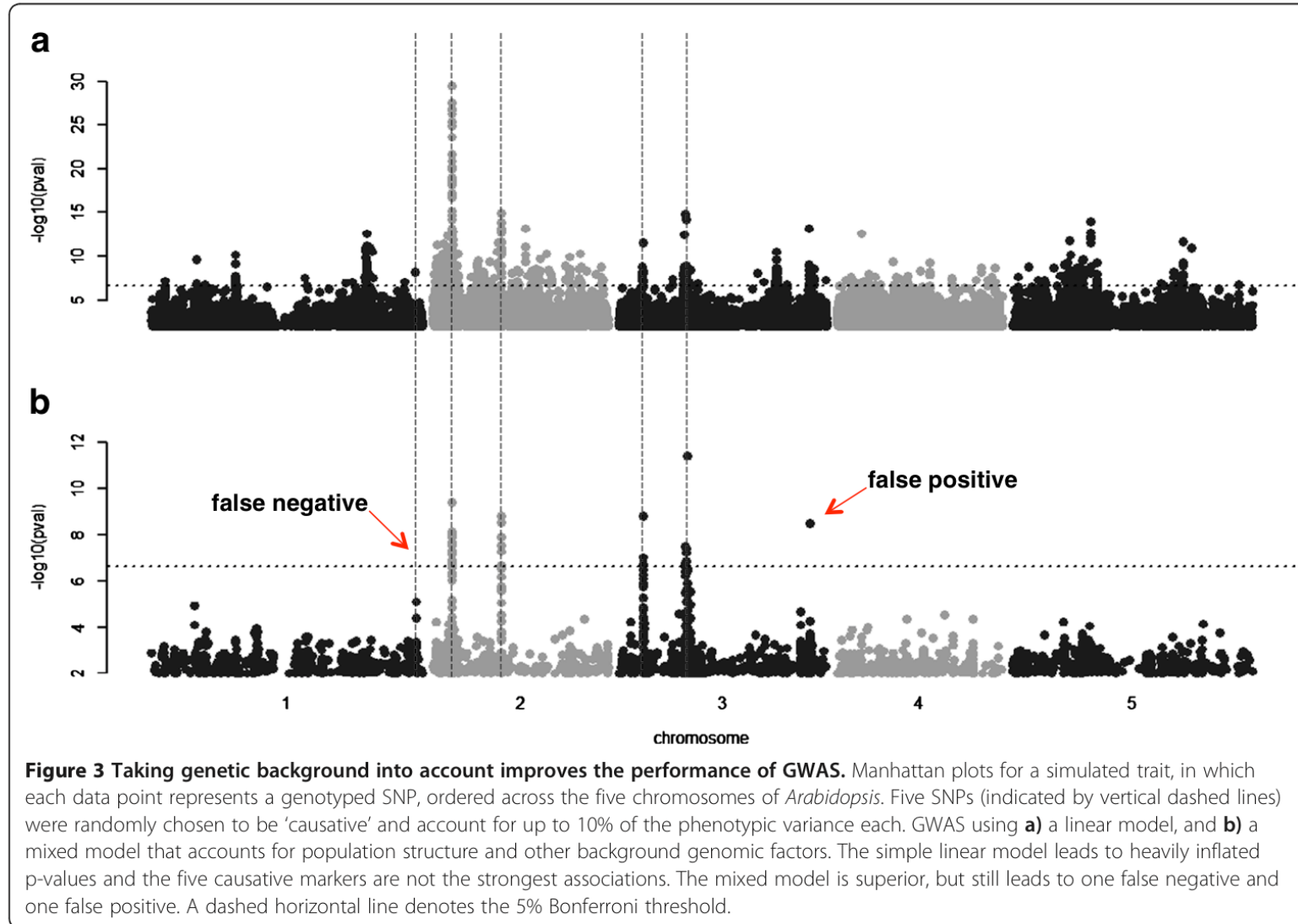
ideal



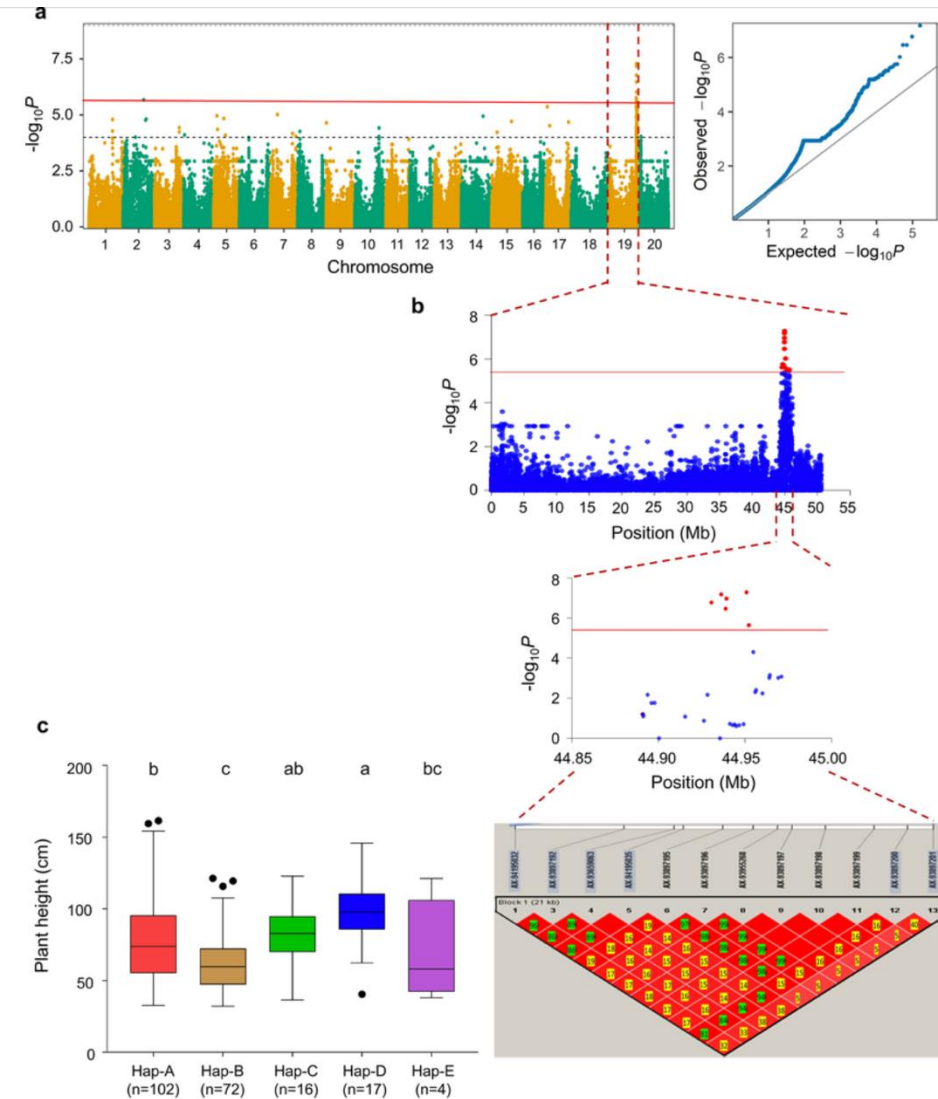
Q-Q plots by four statistical GWAS models



Manhattan plots

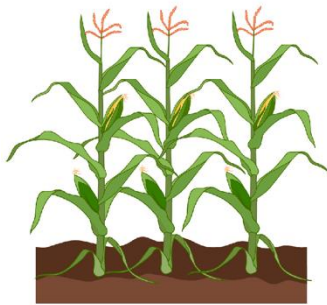


Haplotype Block Analysis

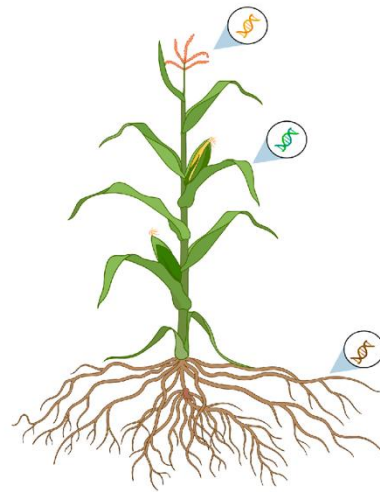


GWAS to Functional Validation

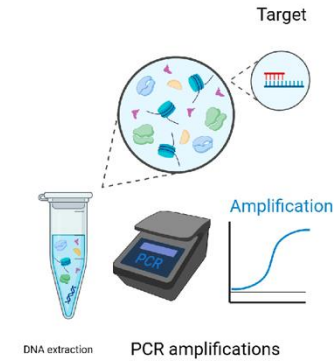
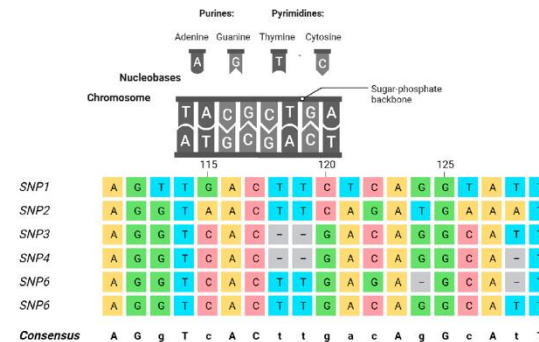
① Collection of diverse population



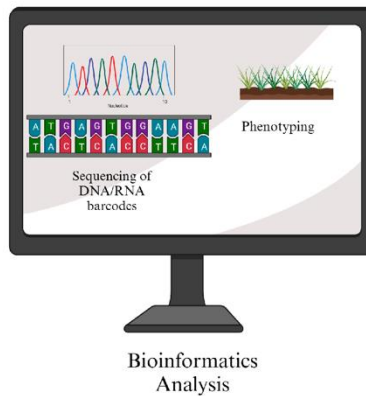
② Phenotyping



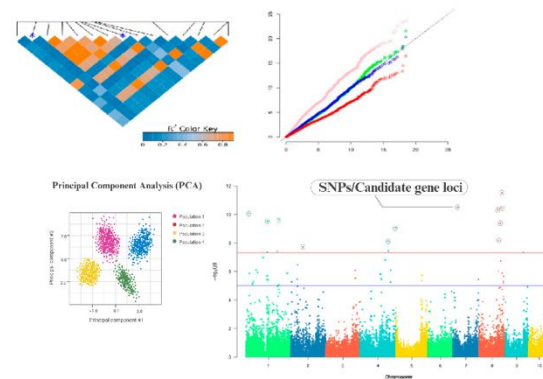
③ Genotypic data (SNPs Q matrix and Kinship matrix (K))



④ Statistical models (GLM, MLM, EMMAX, EMMA)



⑤ GWAS Analysis



⑥ Functional validation

